Lecture 30: Transistor-Transistor Logic

Transistor-Transistor Logic Families

The transistor-transistor logic families are entirely made with bipolar junction transistors (BJTs) and resistors.

But why are they named ‘transistor-transistor logic’ and not simply ‘transistor logic’?That’s because BJTs are used as the main components in both the logic system of the circuit as well as the amplification part of the circuit.

Commercially, the TTL logic components begin with ‘74XX’, such as 7404, 74S86, etc.

For logic gates built using TTL components, the input voltage is applied to the emitter of the transistors. Circuits comprising of the TTL families usually consist of multi-emitter transistors, i.e., multiple parallel transistors with distinct emitter inputs and common base and collector terminals.

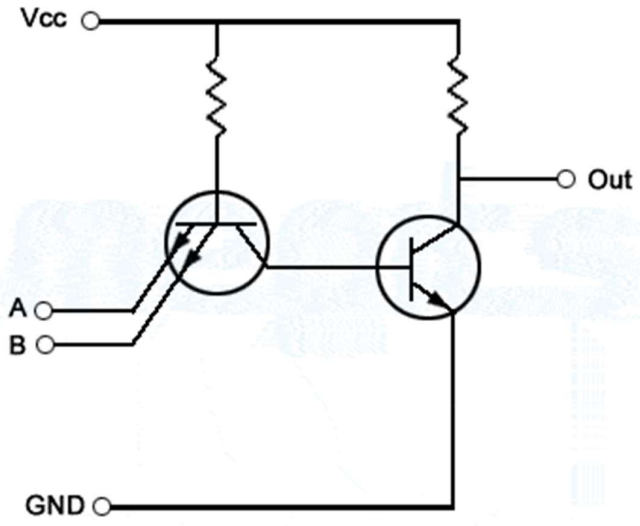


Fig: basic two input TTL NAND gate

The transistors A and B in Fig above have separate inputs at the transistors, but the bases are made common and biased with a resistor, and the collector terminals have also been combined, acting as the base of the next transistor.

In the given 2-input TTL configuration, the application of a logic high voltage at terminals A and B reverse biases the emitter-base junction of the transistor. Thus, a small current flows in the opposite direction. This small current reaches the base terminal of the output, turning on the transistor. Upon switching on, the transistor pulls down its output to a logic low. Conversely, if either of the inputs at terminals A and B is zero, no current reaches the base of the second transistor. Thus the output becomes V_{CC}.

Apart from using normal transistors, Schottky transistors are also preferred for TTL logic systems. These transistors portray the Schottky effect and thus have higher switching speed and lesser power consumption.

TTL is now considered a little outdated, but is used as switching device is relays and driving lamps, and in printers and video display terminals.

Schottky effect: Conduction of current happens with the excitation of electrons from the surface of conducting material, and this excitation happens due to the application of a voltage or due to thermal excitement. The minimum energy required to excite an electron is known as the work function. If we increase the external excitation, the electrons start escaping the surface of the material at a faster rate. If this keeps increasing, the work function slowly decreases. At very high values of the external field, the electron emission increases excessively. This phenomenon is known as the Schottky effect.

Characteristics of TTL Logic Families

- The output of a TTL device can serve as an input to a maximum of 10 gates, i.e., the fan-out is 10**.**
- A logic low voltage for a TTL is defined between 0V to 0.8V.
- A logic high voltage for a TTL is from 2V to 5V.
- The noise margin is at around 4V.
- The propagation delay is about 9ns**.**
- A typical TTL component draws a power of about 11mW**.**

Advantages of TTL Logic Families

- TTL has a strong drive capability.
- It is least susceptible to electrical damage.
- Requires only one supply voltage (otherwise for CMOS)
- Lesser immune to noise when compared to ECL, but more than CMOS.
- Fastest saturation, when compared to other logic families
- Low output impedance for all states

Disadvantages of TTL Logic Families

- TTL dissipates a lot of power, thus not making it suitable for battery-powered devices.
- Not recommended in VLSI chips as it requires more space and isolation
- Expensive compared to MOSFETs

Common TTL Logic ICs

- 74 family
- 74LS (Low-power Schottky) family
- 74F (Fast) family
- 74AS (Advanced Schottky) family
- 74ALS (Advanced Low-power Schottky) family

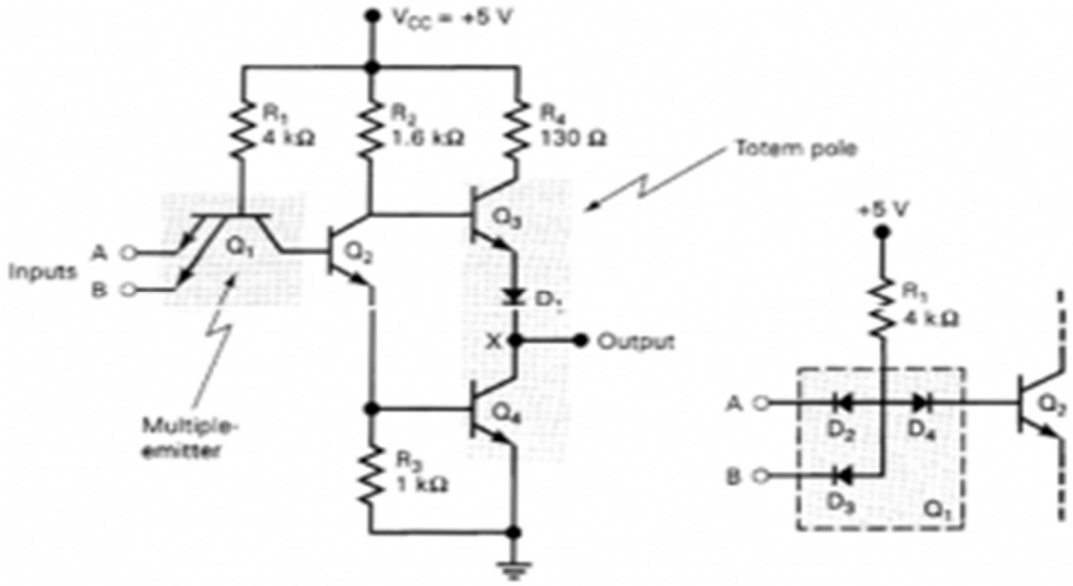
2 input TTL NAND gate operation with Totem pole output

A two input standard TTL NAND gate is a multiple emitter transistor for the inputs A and B. the output transistors Q_3 and Q_4 form a totem-pole output arrangement.

Operation:

If A or B is low or both A and B are low at same time, the base-emitter junction of Q_1 is forward biased and its base-collector junction is reverse biased. Then there is a current from V_{cc} through R_1 to the base emitter junction of Q_1 and into the LOW input, which provides a path to the ground for the current. Hence there is no current into the base of Q_2 and making it into cur-off. The collector of Q_2 is HIGH and turns Q_3 into saturation. Since Q_3 acts as a emitter follower, by providing a low impedance path from V_{cc} to the output, making the output into HIGH.[so for A=B=0 or any of A or B is 0 then output Y=1] At the same time, the emitter of Q_2 is at ground potential, keeping Q_4 OFF.

When A and B are high, the two input base emitter junctions of Q_1 are reverse biased and its base collector junction is forward biased. This permits current through R_1 and the base collector junction of Q_1 into the base of Q_2, thus driving Q_2 into saturation. As a result Q_4 is turned ON by Q_2, and producing LOW output which is near ground potential[so for A=B=1 the output Y=0]. At the same time, the collector of Q_2 is sufficiently at LOW voltage level to keep Q_3 OFF.[so the circuit behaves like a NAND gate]



TTL outputs: Totem pole/ active pull-up

It is possible in TTL gates the charging of output capacitance without corresponding increase in power dissipation with the help of an output circuit arrangement referred to as an active pull-up or totem-pole output. In this case,

- Outputs must never be connected together.
- Connecting outputs causes excessively high currents to flow.
- Outputs will eventually be damaged.
- The standard TTL output configuration with a HIGH output and a LOW output transistor, only one of which is active at any time.
- A phase splitter transistor controls which transistor is active.

One of the problems with the TTL gate circuit is that the pull-up resistor on the output transistor will prevent rapid charging of any wiring capacitance on the output. One way to improve the rise time is to reduce the resistance value as is often done, but this also increases the power dissipation when the output is low. If we look at the circuit, we observe that when the transistor is saturated, it presents a very low effective resistance to ground. The problem arises when the output is high and the pull-up resistor is too large. Ideally we would like to have a very low resistance pull-up when the output is high, but a very high pull-up resistance when the output is low. In this way, we could get quick charging and very low power dissipation.

----------END OF LECTURE----------

REFERENCES

1. **Text book**: **T1.** Modern Digital Electronics by R. P. Jain (Tata McGraw Hill)
2**. Reference book**: **R1.** Fundamental of digital circuits by A. Anand Kumar (PHI)

